Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/28/2005
PI Name:	Lee, Jerry Ssu-Hsien Ph.D.		
Project Title:	Investigating the effects of polarity on protein expression and post-translational modification in cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21218	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Cell polarity mediates intracellular organelle arrangement through the microtubule-organizing center (MTOC). Loss of microtubule structure and the MTOC has been shown to occur in cells during space launch, and although both return during spaceflight, the cells exhibit growth arrest and increased apoptosis. MTOC repositioning causes reorientation of the Golgi towards the anterior of the cell, which may result in changes in protein expression and post- translation modifications. Using a novel mechanical shear assay, I have demonstrated that Swiss 3T3 fibroblasts polarize in the direction of flow in a Cdc42 dependent manner. Thus, I propose to study the effects of cell polarity, and subsequent organelle reposition, on protein expression and modification in cells. This would provide insight into understanding, and overcoming, the short and long term effects of space flight on humans.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	Cells in vivo are constantly subjected to mechanical shear stresses that play important regulatory roles in various physiological and pathological processes. Cytoskeletal reorganizations that take place in response to shear flows have been studied extensively, but whether and how the cytoplasm of cells adapts their mechanical properties to respond to shear remains largely unknown. Here we develop a new method where fluorescent nanoparticles are ballistically injected into the cells to probe with high resolution the local viscoelastic properties of the cytoplasm of individual cells subjected to shear flow. This new assay, ballistic intracellular nanorheology (BIN), reveals that shear flow induces a dramatic sustained 25-fold increase in cytoplasmic viscosity in serum-starved Swiss 3T3 fibroblasts. In contrast, cells stimulated with LPA show highly transient stiffening of much lower amplitude, despite similar cytoskeletal formations. Shear-induced cytoplasmic stiffening is attenuated by inhibiting actomyosin interactions and entirely eliminated by specific Rho-kinase inhibition. Together these results show that biochemical and biophysical stimuli may elicit the formation of qualitatively similar cytoskeleton structures (including stress fibers and focal adhesions), but induces quantitatively different micromechanical responses. Our results suggest that when cells are subjected to shear stresses, its first order of action is to prevent detachment from its substratum by greatly stiffening its cytoplasm through enhanced actin assembly and Rho-kinase mediated contractility.
Bibliography Type:	Description: (Last Updated: 09/01/2009)
Articles in Peer-reviewed Journals	Tseng Y, Kole TP, Lee JS, Fedorov E, Almo SC, Schafer BW, Wirtz D. "How actin crosslinking and bundling proteins cooperate to generate an enhanced cell mechanical response." Biochem Biophys Res Commun. 2005 Aug 19;334(1):183-92. PMID: 15992772 , Aug-2005
Articles in Peer-reviewed Journals	Lee JS, Chang MI, Tseng Y, Wirtz D. "Cdc42 mediates nucleus movement and MTOC polarization in Swiss 3T3 fibroblasts under mechanical shear stress." Mol Biol Cell. 2005 Feb;16(2):871-80. PMID: 15548602 , Feb-2005
Presentation	Lee, J.S.H., P. Panorchan, Y. Tseng, and D. Wirtz "Intracellular Response to Shear Flow is Mediated by Rho-kinase" American Society for Cell Biology Annual Meeting, San Francisco, CA Dec-2005
Presentation	Lee, J.S.H., M.I. Chang, Y. Tseng, and D. Wirtz "Nucleus Movement of Shear Polarized Fibroblasts is Mediated by Cdc42 and Microtubules" American Society for Cell Biology Annual Meeting, San Francisco, CA Dec-2005
Presentation	P. Panorchan, J.S.H. Lee, Y. Tseng, and D. Wirtz. "Taking Cell Mechanics to the Third Dimension" American Society for Cell Biology Annual Meeting, San Francisco, CA Dec-2005
Presentation	Lee, J.S.H., P. Panorchan, Y. Tseng, and D. Wirtz. "Nuclear microrheology" The Society of Rheology Annual Meeting, Vancouver, Canada Oct-2005
Presentation	Tseng, Y., T.P. Kole, J.S.H. Lee, P. Panorchan, and D. Wirtz. "Live-cell Microrheology" The Society of Rheology Annual Meeting, Vancouver, Canada Oct-2005
Presentation	Lee, J.S.H., P. Panorchan, Y. Tseng, and D. Wirtz. "Micromechanical Response of Fibroblasts to Shear Stimulus is Regulated via Rho Signaling Pathway" Biomedical Engineering Society Annual Meeting, Baltimore, MD Sep-2005
Presentation	Panorchan, P., J.S.H. Lee, Y. Tseng, and D. Wirtz. "Extracellular space dimensionality determines intracellular mechanics" Biomedical Engineering Society Annual Meeting, Baltimore, MD Sep-2005